

MAT282 · FALL 2026 · PRACTICE ANSWER KEY

Answer key

Form P · Practice

Worked solutions and marking guide · 50 points · may be shared with students after the review

Read this first. This practice trains the same skills as the real midterm through different questions and a different kind of result (no outlier, a moderate relationship). Aim to understand why each step works. Apply follow-through. Exact values: $\bar{y} = 119.0$; $S_{xx} = 58.38$; $S_{xy} = 196.10$; $S_{yy} = 2472$; $b = 3.359$; $a = 95.82$.

1 · Quick answer sheet

Q	Expected answer	Pts	How the marks split
A1	Grid (rows: direction, days, testable): (i) $\times \checkmark \times$ · (ii) $\checkmark \checkmark \checkmark$ · (iii) $\times \times \times$ (people, not days; not measurable). Rewrite (i) or (iii): direction + days + checkable variable.	2	1 grid (≥ 7 of 9) + 1 rewrite
A2	A supports · B falsifies · C falsifies (no relationship) · D unclear: curved pattern, r would mislead; plot by ranges / add variables.	2	1 (≥ 3 of 4) + 1 next step
A3	1 reliability / comparability · 2 validity (units sold $\neq$ enjoyment) · 3 reliability / comparability (different sensors). One correct explanation.	2	1 (≥ 2 of 3) + 1 explanation
B1	Mean = $1190 \div 10 = 119.0$; median = 119.5; mode: none.	2	1 mean + median; 1 mode
B2	$\Sigma = 2472$; variance = 274.7; $s = 16.6$.	3	1 each (follow-through)
B3	Sun ≥ 8 h: $n = 4$, mean 125.5; Sun < 8 h: $n = 6$, mean 114.7.	2	1 counts + 1 means
B4	$0.5 \times 131 + 0.3 \times 124 + 0.2 \times 118 = 126.3$; plain mean = 124.3; weighted is higher because the latest (largest) Saturday has most weight.	2	1 + 1
B5	$z(\max 146) = 1.63$; $z(\min 91) = -1.69$: neither $ z > 2 \rightarrow$ no outlier. New day: $z = (195 - 119.0) \div 16.6 = 4.59 \rightarrow z > 2$: investigate (flag, check for error/event, do not just delete).	2	1 z-scores and conclusion; 1 new day
C1	$\times$ at (6.9, 119); line positive, balanced; 10 dots within a minor grid square.	3	1 dots; 1 $\times$; 1 line
C2	$b = 196.10 \div 58.38 = 3.36$; $a = 95.8$; $\hat{y} = 95.8 + 3.36x$; about 3.4 more iced coffees per extra hour of sun.	3	1 + 1 + 1
C3	$r = 0.52$, $R^2 = 0.27$. r is a strength measure, not a % of days; R^2 says about 27% of the variation in units is associated with sunshine.	2	1 + 1
C4	Apr 21: $\hat{y} 114.0$, res -3.0 ; Jul 12: $\hat{y} 125.1$, res 20.9; Jul 28: $\hat{y} 123.7$, res -20.7 ; Aug 12: $\hat{y} 109.9$, res 12.1. MAE = 14.2. Good model: residuals scatter randomly around 0 with no pattern or funnel.	4	1 predictions + 1 residuals + 1 MAE + 1 plot and description
D1	$H2 = \text{COUNTIF}(B2:B11, ">=8")$ · $H3 = \text{AVERAGEIF}(D2:D11, 1, E2:E11)$ · $H4 = \text{CORREL}(B2:B11, E2:E11)$	3	1 each
D2	(a) FORECAST arguments are x, known_y, known_x: the ranges are swapped. (b) $=\text{FORECAST}(8, E2:E11, B2:B11)$.	2	1 + 1
D3	(a) Col 2 = sunshine_hours = 3.23 (Row 1 reversed: weekend, post_change, sunshine, intercept). (b) $t(\text{sun}) = 3.23 \div 0.47 = 6.85$; $t(\text{post}) = 16.96 \div 2.33 = 7.27$; both significant, post_change slightly stronger. (c) After the policy change, daily units are ≈ 17.0 higher, holding sunshine and weekend fixed.	3	1 + 1 + 1
D4	R^2 rises from 0.28 to 0.56; SE of y falls from 17.9 to 14.0: residuals are typically about 3.9 units smaller, so forecasts are a little tighter.	2	1 + 1
D5	Predicted = 135.2; residual = $144 - 135.2 = 8.8$; $ \text{residual} < \text{SE of } y (14.0) \rightarrow$ ordinary day for the model.	2	1 + 1
E1	SE = 2.71; $t = 8.51 \rightarrow$ significant. Too strong: observational before/after; summer weather also changed; statistical significance is not size or cause.	3	1 SE + t; 1 + 1 reasons
E2	Stock 150: $0.3 \times 120 + 0.7 \times 20 = 50$. Stock 180: $0.1 \times 120 + 0.9 \times 45 = 52.5$. Choose 150 (lower expected loss); accept 180 if justified by avoiding stock-outs.	2	1 calculations + 1 choice
F1	Data explanation (noisy / weak measure / sampling); real-world mechanism (customers switch from pastries to cold items in heat); feasible Sheets test (add temperature to LINEST for pastry units, check sign and t); falsifier. ($r = -0.26$)	4	1 each

2 · Worked solutions

#	Date	X	Weekend	Event	Y	Y – mean	(Y – mean) ²
1	Apr 21	5.4	0	0	111	-8.0	64.0

2	May 7	3.2	0	0	91	-28.0	784.0
3	Jun 1	5.9	1	0	107	-12.0	144.0
4	Jul 12	8.7	1	0	146	27.0	729.0
5	Jul 13	11.2	1	0	136	17.0	289.0
6	Jul 16	6.4	0	0	126	7.0	49.0
7	Jul 28	8.3	0	0	103	-16.0	256.0
8	Jul 29	9.9	0	1	117	-2.0	4.0
9	Aug 3	5.8	1	1	131	12.0	144.0
10	Aug 12	4.2	0	1	122	3.0	9.0
	Total				1190	0.0	2472.0

